Project Specification – Sustainable Lifestyle Calculator

Project Overview

The Sustainable Lifestyle Calculator is a Python-based application that helps users understand the environmental impact of their everyday habits. Many people want to live more sustainably but are unsure how activities such as transportation choices, electricity consumption, water usage, recycling habits, and plastic usage affect the environment.

This project converts daily lifestyle habits into a measurable sustainability score. The application collects user data, calculates category-level sustainability scores, combines them into an overall Eco Score, and provides personalized feedback and sustainability recommendations.

The goal is to encourage users to become more aware of their environmental impact and motivate small but meaningful lifestyle improvements.

This project aligns with my personal interest in sustainability, waste reduction, composting, and eco-friendly living. It combines environmental awareness with interactive software design.

Use Case

A user wants to understand how environmentally friendly their habits are. They open the Sustainable Lifestyle Calculator and enter information about:

- Distance traveled by car, motorcycle, bus, bicycle, and walking
- Electricity consumption
- Water consumption
- Recycling habits
- Single-use plastic usage

The calculator analyzes this data using sustainability scores and category weights.

The system generates:

- Category scores
- Overall Eco Score (0–100)
- Achievement level
- Personalized feedback

Example:

“You achieved Eco-Friendly status. Your recycling habits positively affected your score. Reducing single-use plastic usage could improve your score further.”

Users and Stakeholders

Primary Users

- Students
- Families
- Sustainability-conscious individuals
- People interested in improving eco-friendly habits

Secondary Stakeholders

- Environmental clubs
- Schools
- Community sustainability programs
- NGOs promoting environmental awareness

Problem It Solves

Many people want to live sustainably but lack simple tools to understand the environmental impact of daily habits.

Most users do not know how transportation, energy use, water consumption, recycling, and plastic waste affect sustainability.

This application addresses that gap by converting sustainability-related activities into measurable insights and actionable feedback.

Primary Interaction Flow

1. User launches application
2. User selects an option from the main menu
3. User enters sustainability data
4. Application validates input

5. Category scores are calculated
6. Eco Score is generated
7. Achievement level is assigned
8. Personalized feedback is shown
9. Eco Score is saved
10. User may view progress graph

Task Vignette 1 – Entering Sustainability Data

The user inputs:

Transportation

- Car distance
- Motorcycle distance
- Bus distance
- Bicycle distance
- Walking distance

Energy and Resource Usage

- Electricity usage
- Water usage

Waste Management

- Recycling habits
- Plastic usage

Technical Notes

- Inputs collected using Python input()
- Numeric inputs validated
- Invalid values rejected
- Negative values not allowed

Task Vignette 2 – Calculating Sustainability Scores

Each category receives a score based on sustainability impact.

Example transportation impacts:

- Car = -3
- Motorcycle = -2
- Bus = -1
- Bicycle = +1
- Walking = +1.5

Category weights:

- Transportation = 30%
- Electricity = 20%
- Water = 15%
- Recycling = 20%
- Plastic Usage = 15%

Scores are normalized into a final Eco Score from 0–100.

Task Vignette 3 – Viewing Results and Feedback

After calculation, users receive:

- Category scores
- Final Eco Score
- Achievement level
- Recommendations

Achievement Levels:

- 0–20 = Unsustainable
- 21–40 = Beginner
- 41–60 = Improving
- 61–80 = Eco-Friendly
- 81–100 = Sustainability Champion

Feedback is generated using conditional logic.

Example:

“Transportation has a high carbon footprint. Consider walking or cycling more.”

Task Vignette 4 – Viewing Progress Graph

Users can save Eco Scores from multiple sessions and visualize progress over time.

Current implementation supports:

- CSV score storage
- Line graph visualization using matplotlib

Not yet implemented

- Pie charts
- Category-wise trend visualization
- Multi-variable analytics

Interface

Version 1 (Implemented)

- Command Line Interface (CLI)
- Text-based interaction using Python

Future Interface (Nice to Have)

- Streamlit dashboard
- GUI application
- Web-based sustainability dashboard

Data and Processing

Input Data

- Car distance
- Motorcycle distance
- Bus distance
- Bicycle distance
- Walking distance
- Electricity usage
- Water usage
- Recycling habit
- Plastic usage

Processing Flow

System performs:

1. Collect inputs
2. Validate data
3. Calculate category scores
4. Apply weighted scoring
5. Generate Eco Score
6. Assign achievement level
7. Generate feedback
8. Save score to CSV
9. Generate progress graph

Output

Application displays:

- Category scores
- Eco Score
- Achievement level
- Sustainability feedback
- Progress graph

Example:

Eco Score: 76
Achievement Level: Eco-Friendly
Recommendation: Reduce single-use plastic usage and continue sustainable transport habits.

Technical Flow

User Input
↓
Validation
↓
Category Scoring
↓
Weighted Score Calculation
↓

Eco Score Generation
↓
Achievement Classification
↓
Feedback Generation
↓
CSV Storage
↓
Graph Visualization

Python Modules Used

Implemented modules:

- csv
- matplotlib
- os

Removed

- pandas (not used in final implementation)

Data Storage

Current storage file:

`eco_scores.csv`

Current columns:

- score

Future Enhancement

Possible expansion to store:

- date
- category scores
- user input history

Challenges

The biggest challenge was creating a sustainability scoring system that is fair, meaningful, and simple enough to implement.

Additional challenges included:

- Determining category weights
- Designing realistic score thresholds
- Generating useful feedback
- Implementing data persistence

What I Can Do Myself

- UX and interaction design
- Sustainability scoring logic
- CLI application development
- Data handling
- Graph generation
- Future dashboard concepts

Final Self Assessment

The project evolved from a basic sustainability tracker into a scoring and feedback system with progress visualization.

A key learning was that sustainability cannot be accurately represented using a single simple metric. Different lifestyle factors have different environmental impacts, making weighted scoring necessary.

To ensure the project remains feasible, I intentionally simplified certain advanced features such as full dashboard development and complex analytics.

The core application is now functional and demonstrates all major planned concepts. Remaining work focuses on improving analytics, recommendations, and user experience rather than building core functionality from scratch.